AFTER AN ADMISSIBLE BLOWUP OF A COMPACTIFICATION, A MORPHISM OF FINITE TYPE SEPARATED OVER THE BASE EXTENDS TO A PROPER MORPHISM FROM AN OPEN CONTAINING THE ORIGINAL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0F3Z (sources/stacks-flat/pinned-original-source.tex line 9762).

Allowed standard prerequisites (editorial).

- Scheme theoretic image after an admissible blowup is the strict transform of the original image. Same chapter, Lemma labelled lemma-replaced-by-strict-transform (source line 9728), invoked at proof lines 9773-9776. Supplies that the scheme theoretic image of the diagonal morphism is replaced by its strict transform when Y is blown up; it is counted separately in this lane as S12-algebra-replaced-by-strict-transform.
- Zariski after blowup: an admissible blowup making a closed subscheme condition hold. Same chapter, Lemma labelled lemma-zariski-after-blowup (source line 9197), invoked at proof lines 9776-9777. Supplies that after a further admissible blowup the map Z -> Y may be assumed an open immersion.

Source reference. Original lemma 0F3Y: sources/stacks-flat/pinned-original-source.tex, source lines 9728–9728. Same-chapter pointer into the author environment lemma at source line 9728; invoked at proof line 9776 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 081S: sources/stacks-flat/pinned-original-source.tex, source lines 9197–9197. Same-chapter pointer into the author environment lemma at source line 9197; invoked at proof line 9777 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let U be a scheme of finite type and separated over S. Let $V \subset U$ be a quasi-compact open. If V has a compactification $V \subset Y$ over S, then there exists a V-admissible blowing up $Y' \to Y$ and an open $V \subset V' \subset Y'$ such that $V \to U$ extends to a proper morphism $V' \to U$.*

Proof. Consider the scheme theoretic image $Z \subset Y \times_S U$ of the "diagonal" morphism $V \to Y \times_S U$. If we replace Y by a V-admissible blowing up, then Z is replaced by the strict transform with respect to this blowing up, see Lemma 0F3Y. Hence by Lemma 081S we may assume $Z \to Y$ is an open immersion. If $V' \subset Y$ denotes the image, then we see that the induced morphism $V' \to U$ is proper because the projection $Y \times_S U \to U$ is proper and $V' \cong Z$ is a closed subscheme of $Y \times_S U$. □